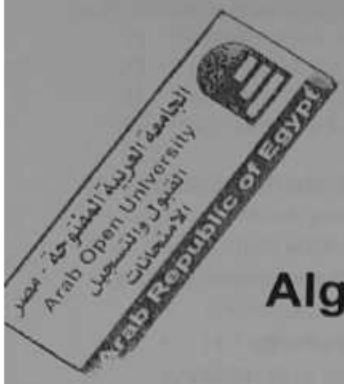


AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M269
Algorithms, Data Structures and Computability

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 14 /April/2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 hours

Instructions:

- Total Marks: **60 marks**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer Booklet.
- Be sure you write your **name and ID** on the External Answer Booklet.
- **Calculators** are not allowed.

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PART 1: ALL QUESTIONS ARE REQUIRED [10 Marks]

Question 1: (10 marks)

For the following statements please answer "True" or "False":

1) MergeSort is not an example of a brute-force strategy.
2) The function $3n^2$ has lower time complexity than $2n^3$.
3) Finding the smallest element in a linked list of size *n* does not run in O(log n) time.
4) Quick Sort has the same worst-case time complexity as Bubble Sort.
5) When analyzing algorithm efficiency, the space factor is not measured by the average memory used.
6) Insertion Sort and Selection Sort exhibit the same time complexity.
7) push() and pop() are not the primary operations for inserting and deleting elements in a Queue.
8) Bubble does not require more memory than Merge Sort.
9) A linked list is not always the optimal choice when data size and structure change frequently.
10) Computing the average of the elements in a stack of size *n* is not an O(1) operation.

PART 2: ALL QUESTIONS ARE REQUIRED [30 Marks]

Question 2: (16 marks)

a) Identify each of the following descriptions in a couple of words: (4 marks)
   1) The process of examining a problem using the analytical methods, techniques, and computational skills provided by the field of computer science.
   2) A problem articulated in a form that allows for the development and construction of an algorithmic solution.

b) Let A be an array with N integer numbers (4 marks)
   1) What's the sorting algorithm that should be used to achieve your choice?
   2) What is complexity of the fastest algorithm to sort them?

c) Arrange the complexities of the following operations (from best to worst): (4 marks)

   4^n , 4 log n, 4 log 4, 4 n, $4n^4$, n log n

d) For each of the following scenarios select the "best" data structure that should be used:
   (4 marks: 1 mark each)
   1) The undo feature available in a web browser.
   2) A representation of an organization's hierarchical authority structure.
   3) A dataset whose values are infrequently modified.
   4) An ATM system that serves customers on a first-come, first-served basis.

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Question 3: (14 marks)

a) What is the complexity of the following tasks? (4 marks: 1 mark each)
1) Sorting a list of N elements using Selection-Sort algorithm.
2) Finding the smallest element in the stack of size n.
3) Inserting an item in a queue data-structure.
4) Remove the first item entered in a stack data-structure of size n.

b) For each of the below plots determine their behavior from the following list. (4 marks: 1 mark each)

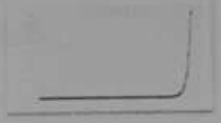
- Cubic ≈ n^3
- Linear ≈ n
- Exponential ≈ 2^n
- N-Log-N ≈ n log n
- Constant ≈ 1
- Quadratic ≈ n^2
- Logarithmic ≈ log n

[Note: The answer should be as follows: for example, Plot X has a Cubic behavior.]

Plot a



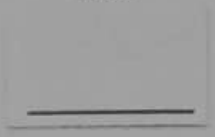
Plot b



Plot c



Plot d



c) Given the following array A [7,12,4,9,19,2,6,8,11,15,20,5,3], answer the following set of questions:
1) Draw the heap presentation of this array (2 marks)
2) What is the depth of node 19? (1 marks)
3) Find the sibling(s) node(s) of node 20. (1 marks)
4) Find the pre-order traversal of this tree. (2 marks)

PART 3: ALL QUESTIONS ARE REQUIRED [20 marks]

Question 4: (10 marks)

a) Find the complexity of the following items, where Arr is an array, n is the input size: [Note: your answer must show the steps that lead to your final answer]

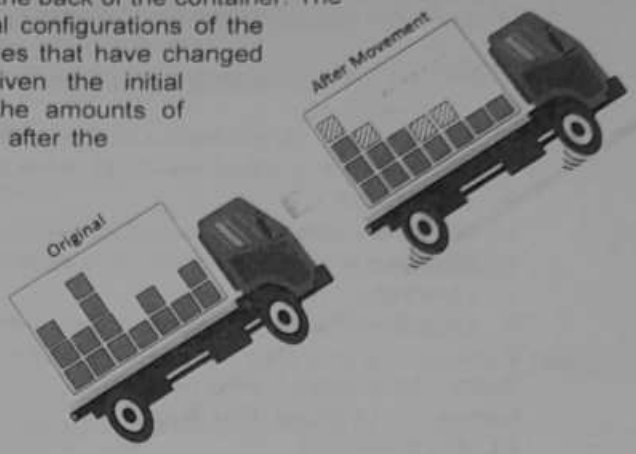
a)	b)	c)
val = 1 for i = 1 to n val = val × i end for for k = 1 to n Arr[k] = Arr[k] + val end for HeapSort(Arr[])	T(n) = 16T(n/4) + n for n > 1 with T(1) = 0	T(n) = T(n-1) + 5 for n > 1 with T(1) = 0

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Question 5: (10 marks)

In a big container, there are n columns of boxes (all of the same size). In the container, the boxes are arranged in columns. At first, the boxes are randomly aligned over each other's. When the container moves forward, it begins to pull all the boxes to the back of the container. The figure shows the initial and final configurations of the boxes in the container: the boxes that have changed their position are hashed. Given the initial alignment of the boxes, find the amounts of boxes in each of the n columns after the movement.



	Input	Output	Note
Test case 1	[3,2,4,1,2,1,2]	[4,3,2,2,2,1,1]	As illustrated in the figure
Test case 2	[7,6,5,3,2,1]	[7,6,5,3,2,1]	No change

Answer the following questions:

a) Design an algorithm to solve this problem (6 marks)
b) Derive the complexity of your algorithm. (4 marks)

End of questions



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